Predicting *Mental Health Trends*

GROUP 11 -

Aman Ostwal (ago34@drexel.edu)

Darshit Rai (dr3264@drexel.edu)

Sai Pokuri (sp3867@drexel.edu)

Sanjoli Sogani (ss5656@drexel.edu)

Introduction

- Mental health is a growing concern worldwide, with increasing demand for proactive interventions. By leveraging data science and machine learning, we can analyze patterns in mental health trends, identify risk factors, and forecast future needs. This presentation explores how predictive analytics can enhance early detection, optimize mental health resources, and support data-driven decision-making.
- **Why It Matters:** Rising mental health challenges require innovative solutions.
- **Our Approach:** Using data-driven models to predict trends and inform interventions.
- **Key Benefits:** Improved early detection, better resource allocation, and enhanced support systems.

Goal

1. Leverage Reddit discussions to analyze mental health trends and identify emerging concerns.
2. Collect and preprocess data from mental health-related subreddits.
3. Classify discussions into severity levels (high, medium, low) using an evidence-based algorithm.
4. Track sentiment fluctuations over time to detect patterns in community well-being.
5. Apply machine learning techniques to predict future mental health trends.
6. Enhance awareness and support early intervention through data-driven insights.

Data Acquisition: Overview & Methodology

- **Objective:** Collect real-time data from Reddit for mental health analysis
- **Target Subreddits:** r/anxiety, r/depression, r/adhd
- **Approach:**
  - Utilize the PRAW library with required API credentials (client id, client secret, user agent)
  - Batch fetching (up to 10,000 posts per batch) until posts older than 10 days are reached
- **Outcome:** Establish a robust, real-time dataset with key metadata

Author	Created_UTC	Created_DateTime	Score	Selftext	Tokenized_Selftext	Subreddit	Title
Flashy_Highway_8937	1739053989	2/8/2025 22:33	2	hi like many i suffer from bad healt	['hi', '.', 'like', 'many', 'suffer',	anxiety	chest strain or heart issue
Timdillonfanclub	1739053782	2/8/2025 22:29	1	i was prescribed propranolol for m	['prescribed', 'propranolol', 'a	anxiety	trying propranolol
user12747	1739053495	2/8/2025 22:24	1	our house is 1970s and its very like	['house', '1970s', 'â€™', 'likely	anxiety	super worried about asbestos exposure
Nique_T	1739053377	2/8/2025 22:22	2	is anyone waking up feeling weak (	['anyone', 'waking', 'feeling',	anxiety	is anyone
scubasteve_XYZ	1739053336	2/8/2025 22:22	1	i am going nuts my boyfriend and i	['going', 'nuts', '.', 'boyfriend'	anxiety	sometimes im okay and then shanties i remember im literally moving c
comet_lobster	1739053226	2/8/2025 22:20	1	not sure if this counts as a trigger f	['sure', 'counts', 'trigger', 'any	anxiety	i hate how i always get anxiety in the evenings
AlwaysTheWrongGirl	1739052806	2/8/2025 22:13	1	hello i am in a ldr right now but we	['hello', ',', 'ldr', 'right', 'meet	anxiety	my bf doesnt recognize or remember me helping him through his anxie
Tjthegod01	1739051747	2/8/2025 21:55	1	my blood work was done 6 month:	['blood', 'work', 'done', '6', 'm	anxiety	health anxiety
beenbetterhbu	1739051114	2/8/2025 21:45	2	a good friend of mine struggles wi	['good', 'friend', 'mine', 'strug	anxiety	my friends anxiety is ruining her life
ilikechips1858	1739050787	2/8/2025 21:39	1	15 please dm	['15', 'please', 'dm']	anxiety	need urgent help
princessgrumru	1739050169	2/8/2025 21:29	2	i always have so much anxiety surr	['always', 'much', 'anxiety', 's	anxiety	college students how do u deal w teststudying anxiety
dunderhead1	1739049976	2/8/2025 21:26	1	about a month ago i had some ligh	['month', 'ago', 'light', 'anxiet	anxiety	has anyone else had health anxiety that turned into general anxiety
Puzzled-Article9621	1739049959	2/8/2025 21:25	1	i am currently having a really bad (	['currently', 'really', 'bad', 'cp	anxiety	how to deal with feeling like your gonna pass out from an anxiety attac
farticus-blarticus	1739049945	2/8/2025 21:25	1	its hard to type my hands are cold	["'s", 'hard', 'type', '.', 'hands'	anxiety	the inner machinations of my mind are an enigma
Silent-Cheesecake475	1739048797	2/8/2025 21:06	2	i had been battling anxiety and de	['battling', 'anxiety', 'depressi	anxiety	anxiety returned 2025

Dataset Description

- Post Title – Quick summary of the discussion
- Selftext – the body of the post, where users express their thoughts in detail.
- Author – to track user engagement and potential behavioral patterns.
- Timestamp – which helps us analyze trends over time.
- Score – which indicates user interactions through upvotes and downvotes.
- Subreddit Tag – to retain the context of where the post originated.

Data Integration & Implementation

Extracted Data Attributes:

Post Title, Selftext, Author, Timestamp, Score

Each record tagged with its originating subreddit

Integration:

Merging data from multiple subreddits into a unified dataset

Preserving key metadata for downstream analysis

Custom Python Script:

API initialization and continuous data fetching loop

Batch processing with error handling

Data Handling:

Integration into a Pandas DataFrame

Exporting the raw data as a CSV file

Code Highlights

```python
import pandas as pd

data = []
subreddits = ["anxiety", "depression", "adhd"]
for subreddit_name in subreddits:
    subreddit = reddit.subreddit(subreddit_name)
    for post in subreddit.new(limit=10000):
        data.append({
            "Title": post.title,
            "Content": post.selftext,
            "Subreddit": subreddit_name,
            "Created_UTC": post.created_utc
        })

# Convert to DataFrame
df = pd.DataFrame(data)
```

```python
import praw

# Initialize Reddit API
reddit = praw.Reddit(
    client_id="YOUR_CLIENT_ID",
    client_secret="YOUR_CLIENT_SECRET",
    user_agent="YOUR_USER_AGENT"
)

subreddits = ["anxiety", "depression", "adhd"]
for subreddit_name in subreddits:
    subreddit = reddit.subreddit(subreddit_name)
    for post in subreddit.new(limit=10000):
        print(post.title, post.selftext)
```

```python
df.drop_duplicates(subset=["Author", "Created_UTC", "Title"], inplace=True)
```

```python
output_file = "reddit_posts.csv"
df.to_csv(output_file, index=False)
print(f"Data saved to {output_file}")
```

Challenges, Limitations & Next Steps

Challenges & Limitations:

Reddit API does not provide access to historical data due to recent policy changes

PRAW API is limited to retrieving only the past 10–15 days of data

Mitigation Strategies:

Optimizing data fetch cycles to capture the freshest data available

Monitoring API changes to adapt the data acquisition process

Next Steps:

Focus on advanced analysis using the integrated dataset

Expand to additional data sources if needed

Data Preprocessing

1. **Handling Missing Values** - Checked for missing values in key columns: Author, Created_UTC, Title, Selftext, Subreddit. Dropped rows with missing essential data. Filled text-based columns with an empty string ("") where appropriate.
2. **Data Type Conversion** - Ensured Created_UTC remained in UNIX timestamp format. Converted Created_DateTime to a proper datetime object. Ensured Score was stored as an integer.
3. **Removing Duplicates** - Identified and removed duplicate rows to maintain data integrity.
4. **Normalizing Text Data** - Converted text columns (Title, Selftext, Tokenized_Selftext) to lowercase. Removed special characters, excessive spaces, and unnecessary punctuation from Title and Selftext. Ensured Tokenized_Selftext was correctly formatted as a list of words.
5. **Handling Outliers** - Checked for anomalies in the Score column (extremely high or low values).
6. **Standardizing the Subreddit Column** - Removed leading/trailing spaces. Ensured consistent capitalization across subreddit names.
7. **Removing Stopwords** - Removed common stopwords (e.g., "the," "is," "and") from Tokenized_Selftext to improve text analysis.
8. **Saving Cleaned Data** - Saved the preprocessed dataset as cleaned_data.csv to preserve the original dataset.

Predictive Modeling & Hybrid Learning Approach

Goal: Classify mental health discussions into structured categories which we show multi-level classification

- **Severity Classification**: Train models to classify posts into High, Medium, or Low Severity
- **Topic-Specific Classification**: Identify the primary mental health disorder discussed (Anxiety, Depression, ADHD)

ML Models Used:

- Random Forest: Handles non-linear classification with feature importance insights.
- XGBoost: Provides high accuracy, especially for imbalanced datasets.

Hyperparameter Tuning:

- Grid Search: Systematic exploration of optimal parameters.
- Random Search: Efficient sampling of hyperparameter space.

Predictive Modeling & Hybrid Learning Approach

Goal**:** Extract hidden themes and group similar mental health discussions.

- **Latent Dirichlet Allocation (LDA)**: Identifies underlying topics within mental health discourse.
- **Clustering Techniques (K-Means)**: Groups posts based on textual and sentiment patterns.

- Target Variables
  - **Mental Health Disorder Type**
    - Classifies the disorder discussed in the post (**Anxiety, Depression, ADHD)**
  - **Severity Level**
    - Rates the **intensity of distress** as **High, Medium, or Low**, based on post content.

Use of Diagnosis Assessment in Modelling

- **PHQ-9 (Depression Assessment)**
  - Standardized **nine-item** questionnaire for detecting and measuring **depression severity and** classifies depression into **Minimal, Mild, Moderate, and Severe** levels.
- **GAD-7 (Anxiety Assessment)**
  - **Seven-item** scale assessing **generalized anxiety disorder** severity and used for clinical screening and symptom tracking over time.

DSM-5 (ADHD)

- **Diagnostic manual** for ADHD, Anxiety, Depression, and other mental health conditions and defines **clinical criteria** for accurate diagnosis.

Sentiment and contextual analysis from PHQ-9, GAD-7, and DSM-5 improve early detection of mental health conditions, enabling timely intervention.

Multi-modal learning integrates textual data with additional behavioral indicators (e.g., engagement patterns, response styles) to refine predictive accuracy

Collaboration With Experts

To ensure the accuracy and reliability of our mental health predictions, we are collaborating with industry experts who bring valuable domain knowledge and insights:

Ashleigh Zoerb – Licensed Mental Health Therapist (10+ years of experience)

- Provides expertise in assessing mental health severity.
- Helps validate the severity classification algorithm based on real-world mental health cases.

Nelca Ece Yilmaz – Masters in Psychology

- Provides expertise in assessing mental health severity.
- Helps validate the severity classification algorithm based on real-world mental health cases.

Their guidance ensures that our model aligns with professional mental health assessments, enhancing the credibility and impact of our research.

THANK YOU!!